



P26

Weighted BFT and Reputation: From Equal-Weight $2f+1$ to Credibility-Weighted Stop Decisions — A Design Direction

加权 BFT 与声誉：等权 $2f+1$ 的权重方向

Weighted BFT and Reputation: From Equal-Weight $2f+1$ to Credibility-Weighted Stop Decisions — A Design Direction

Working Title (EN): Weighted BFT and Reputation: From Equal-Weight $2f+1$ Quorums to Credibility-Weighted Stop Decisions — A Design Direction, Not an Implementation

Consensus governance paper P26 • the weight dimension of v7.5.0 P2 "BFT-lite Stop Decisions" • sister paper to P16 (typed finality) and P31 (fail-safe recovery) • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: The committee of v7.5.0 P2 is a **one-person-one-vote, equal-weight $2f+1$** quorum—among 7 QA everyone's vote weighs the same. This is correct when "all members are of comparable ability and wrong votes are independent"; but in a real LLM committee, member quality is uneven and historical performance is traceable. This paper discusses a **design direction**: upgrading the "equal-weight quorum" into a "quorum weighted by historical credibility," and surveys three recent threads—weighted BFT / coalition governance / proof-of-thought consensus—but explicitly states: **UDOS currently does not implement weighted voting, and all benefits are design hypotheses marked [RESULT NEEDED]**.

Evidence scope (stated only once in the whole paper): All UDOS-side numbers go back to v7.5.0 P2 (cpu-proto / verified). The recent works mentioned in this paper—weighted BFT (WBFT class), CP-WBFT, COALITION-VAST, BlockAgents (Proof-of-Thought)—all come from research threads provided by the user, and **their exact numbers have not been verified one by one in the v7.6.0 evidence ledger (EVIDENCE_LEDGER_v76.md)**; wherever specific percentages/thresholds are cited, they are labeled "thread scope • unverified" or [CITATION NEEDED] and are not treated as facts. The safety of equal-weight $2f+1$ and view change comes from classical PBFT (Castro & Liskov, 1999). Verification date 2026-09-21.

Abstract

v7.5.0 P2 uses equal-weight BFT-lite to guard the "no wrong stop" bottom line of the LLM acceptance committee: $n \geq 3f+1$, quorum $q=2f+1$, equivocation invalidates the whole round, and more than f silence turns to view change. The equal-weight design has a clean safety proof—there are only $n-f \geq 2f+1=q$ honest votes, so it is impossible for STOP and CONTINUE to each gather q . But this proof implicitly assumes: **the f Byzantine members are a "worst-case uniformly distributed minority," and all honest members have equal voting weight.** A real LLM committee does not satisfy the latter: some QA have high historical accuracy, while others often overstep or are ambiguous; counting the votes of these two classes as equally heavy amounts to wasting the free signal of "who is more credible."

This paper surveys the "equal weight $\rightarrow$ weighted" design space along three directions rather than implementing it. First, **weighting by historical credibility** (WBFT-class idea): bind each member's voting weight to its past acceptance accuracy so that unreliable members' votes contribute less to the quorum; the cost is introducing a new attack surface of "how weights update and how to prevent weight manipulation." Second, **coalition governance and targeted audit** (COALITION-VAST-class idea): when Byzantine behavior is not single-point malice but a few members **colluding** to induce agreement, equal-weight majority votes are precisely the target exploited by collusion; such works use coalition game theory + on-chain rule evolution + trust-driven targeted audit to discover "collusive agreement." Third, **turning the thought process into a consensus object** (BlockAgents' Proof-of-Thought): vote not only

on the final label but on the "reasoning process" and stake on it, making poisoning/backdoor attacks harder to hide.

This paper's conclusion is restrained: **weighting and reputation are the natural next dimension of P2's equal-weight design, but it is a design direction rather than an implementation.** The reason the equal-weight $2f+1$ proof is clean is precisely that it does not depend on the correctness of any "weight estimation"; once weights are introduced, safety changes from "pure combinatorial" to the product of two variables, "combinatorial $\times$ weight-estimation quality," and the withdrawal lesson of H-CSC v2 (a learned module being worse under a fair baseline) precisely warns us: do not rush to put the "smarter-looking" weighting logic into a safety-critical path. The benefits of all weighting schemes are [RESULT NEEDED] .

Keywords: weighted BFT; reputation weights; quorum; coalition governance; proof of thought; stop decision; design direction

Structured Abstract (Background problem $\rightarrow$ Argument $\rightarrow$ Evidence $\rightarrow$ Contributions)

- **Background problem:** P2's equal-weight $2f+1$ assumes "all honest members equal-weight, Byzantine uniformly distributed," but real LLM committee member quality is uneven and the signal of "who is more credible" is wasted by the equal-weight design; meanwhile Byzantine behavior may upgrade from single-point malice to **collusion-induced agreement**, with equal-weight majority votes being the target of collusion.
- **Core argument:** Upgrading the equal-weight quorum to historical-credibility weighting is the natural next dimension of P2; but weighting turns "pure combinatorial safety" into "combinatorial $\times$ weight estimation," adding the attack surface of weight updates and weight manipulation, so it should be cautiously explored as a **design direction** rather than immediately replacing the verified equal-weight committee.
- **Evidence threads:** (1) the equal-weight $2f+1$ safety of classical PBFT (Castro & Liskov, 1999); (2) UDOS P2's 10 property tests and 24-order end-to-end at $n=7, f=2$ (cpu-proto / verified); (3) the three recent threads WBFT / CP-WBFT / COALITION-VAST / BlockAgents (**numbers unverified**); (4) the H-CSC v2 self-withdrawal (weighting/aggregation modules must pass a steelman contrast).
- **Contributions:** (i) formalizing the "equal weight $\rightarrow$ weighted" design space and costs; (ii) extending the Byzantine threat from "single-point malice" to "collusion-induced agreement"; (iii) clearly listing the safety-property list of weighting schemes and [RESULT NEEDED] ; (iv) stating UDOS currently does not implement weighting and only registers the direction.

1 Starting Point: The Clean Proof of Equal-Weight $2f+1$ and Its Assumptions

1.1 Why P2 Chose Equal Weight

The reason the safety proof of v7.5.0 P2 can be written as 10 CI-able property tests is that it **deliberately chose the simplest voting model**—equal weight, one person one vote. Its core proof (verified):

$$\text{\$ \$ } \text{honest votes} = n - f \geq 3f+1 - f = 2f+1 = q \text{ \$ \$ }$$

Under $n \geq 3f+1$, there are at most q honest votes, so STOP and CONTINUE **cannot each** gather q honest votes—this is the safety of "at most one decision value." This proof does not depend on any estimate of "who is more accurate"; it is purely combinatorial. P2 actively discloses: its signatures are SHA-256 hash placeholders that do not resist real forgery, and **semantic common-cause faults break through majority votes**.

1.2 What Equal Weight Wastes

Equal weight assumes "all honest members are equally credible." But in a real LLM committee this assumption often does not hold:

- Member A has high historical acceptance accuracy and is never ambiguous;
- Member B often gives "barely pass" and occasionally oversteps to check a work order that should not stop as pass;
- Member C is newly joined with no history.

The equal-weight design counts A's and B's votes as equally heavy. This means: to gather $q=2f+1$, you must "trust enough people," even if some of them are known unreliable. **Weight is a free signal—it tells you "when gathering q , you need not count everyone."** This is precisely the intuitive source of weighted BFT.

But note at the same time why the proof of Sec. 1.1 is clean: it **needs no** estimate of weights. This is a trade-off—exchanging "simplicity of the safety proof" for "the ability to use credibility signals." The rest of this paper computes both sides of this trade-off clearly.

2 Direction One: Weighting by Historical Credibility (WBFT Class)

2.1 Design Intuition

Let member v_i 's historical credibility be $w_i \geq 0$, and change the quorum from "vote count" to "weight sum":

$$\sum_{i \in S} w_i \geq Q, \quad Q = 2f+1 \text{ \textit{weight units (design proposal)}} \quad \square$$

Intuitively, unreliable members' w_i are small, and gathering Q relies mainly on high-credibility members' votes; this way, **under the same n , fewer high-credibility members need to be "convinced," and Byzantine nodes find it harder to manipulate the stop decision**. The recent weighted-BFT thread (called WBFT class in user research, IEEE TCCN direction, **specific algorithm details and numbers not verified in this ledger, [CITATION NEEDED]**) precisely binds voting weights to each LLM's historical response quality and credibility, so as to incentivize reliable behavior and reduce malicious-node impact. Another thread claims CP-WBFT (AAAI 2026 direction) can tolerate a high proportion of faulty nodes—its claims such as "85.7% fault rate" are **user-thread restatements, unverified**, and this paper does not accept them as facts.

Figure 3 Equal-weight quorum to credibility weighting

Figure 3 Comparison of P2 one-person-one-vote (left) and historical-credibility weighting (right); weighting introduces three new attack surfaces: weight update / cold start / equivocation interaction. Conceptual illustration, not measured data.

2.2 The Cost of This Trade-off

Weighting is not free. After introducing w_i , safety changes from "pure combinatorial" to:

$\text{safety} = \text{combinatorial correctness} \times \text{weight-estimation correctness}$

Specifically, three new attack surfaces are added:

1. **Weight-update manipulation:** A Byzantine member can "behave well early to farm weight and act maliciously at the critical moment"—weight is history, and malice can lag.
2. **Cold start:** A new member's w_i is undetermined, either given 0 (cannot participate) or a default value (equal to equal weight).
3. **Interaction of weight and equivocation:** P2's equivocation whole-round invalidation is purely rule-based; after weighting, "does whole-round invalidation zero out its weight contribution" needs redefinition.

UDOS's stance: Weight updates **must use rule predicates** (such as the agreement rate with the committee's final decision within a rolling window), and weights cannot be dynamically estimated with learned models—this comes directly from P32's red line (H-CSC v2 proving the learned encoder is defeated by the rule predicate in adversarial scenarios). Before P2's property-test suite is extended and the steelman contrast runs through, UDOS **does not implement** weighted voting.

3 Direction Two: Collusion Governance and Targeted Audit (COALITION-VAST Class)

3.1 From "Single-Point Malice" to "Collusion-Induced Agreement"

P2's fault model assumes the f Byzantine members are an **independently malicious minority**. But H-CSC v2 and P2's "semantic common-cause fault" warning point to a more dangerous form: **a few members collude to jointly induce the majority to reach a "looks-aligned" label on the same wrong result**. At this point equal-weight majority votes are **precisely the target exploited**—what the colluders must do is not prevent gathering q , but **help gather q** , letting the system complete the wrong result with high confidence.

This is exactly why P2's binary decision is dangerous: it only asks "whether STOP is gathered," not "whether this STOP is honest agreement or induced agreement." P16's three-level typed finality gives an outlet for "whether the semantic core is truly aligned"; but what if the colluders forge even the semantic core into alignment?

3.2 Coalition Game Theory + Targeted Audit

The recent coalition-governance thread (called COALITION-VAST in user research, IEEE 2026-08 direction, **numbers unverified, [CITATION NEEDED]**) targets precisely such "coalition deviation, governance capture, hasty rule change." Its approach: use coalition game theory to identify "which members' votes exhibit collusion patterns," use on-chain rule evolution to prevent governance from being hastily tampered with, and use trust-driven **targeted audit** (auditing only the most suspicious subset rather than full re-audit) to hold audit cost down. The user thread claims that in three scenarios—medical resource allocation, autonomous swarms, and multi-stakeholder finance—alignment is about 0.90 vs a 0.69 baseline and undetected coalition deviation falls by about 98%—**these numbers are thread restatements, not verified in this ledger, and are not treated as facts**.

The lesson for UDOS is not "copying a governance framework" but **a falsifiable design principle**: the audit of stop decisions cannot only look at "whether the quorum is gathered," but must also look at "whether the voter combination of those votes gathering the quorum exhibits collusion features."

Implementing this principle needs P16's \$C_i\$ (semantic-core fingerprint) and P3's Trace hash chain as the base, all currently [RESULT NEEDED] .

4 Direction Three: Turning the Thought Process into a Consensus Object (BlockAgents / Proof-of-Thought)

4.1 Vote Not Only on Labels but on the Reasoning Process

The object of equal-weight BFT voting is the **final label** (STOP/CONTINUE). A deeper thread is: turn the **reasoning process itself** into a consensus object. In user research, BlockAgents (Shanghai Jiao Tong University team, ACM Turing Award Celebration China 2024 direction) is said to propose Proof-of-Thought (PoT) consensus: combining stake-based voter designation with multi-round debate-style voting to hold the accuracy interference of poisoning attacks at a low level and the backdoor success rate at a low level (**user thread claims <3% / <5%, unverified, [CITATION NEEDED]**).

Intuitively, voting on the "reasoning process" is harder to cheat than voting only on the "conclusion"—because the malicious actor must not only forge the conclusion but also forge a "plausible reasoning chain," while each step on the chain can be independently checked. This shares a philosophical origin with P16's semantic_commit (committing to the semantic core rather than only the label): **the finer the commit granularity, the higher the cheating cost, but the higher the aggregation cost as well.**

4.2 Division of Labor with P16

P16 already takes "committing to the semantic core" as one level of three-level finality; BlockAgents-class PoT pushes "committing to each step of the reasoning chain" even finer. UDOS's boundary is: **how fine is worth going?** Doing $2f+1$ endorsement for each reasoning step costs $O(\text{number of steps})$ times fan-in, directly coupled with P1's topology complexity. Before actual measurement proves the wrong-stop-rate improvement of "voting on the reasoning chain" over "voting on the label" is **greater than** its cost, UDOS only registers it as a direction and does not implement it.

5 Contrast and Trade-offs of the Three Directions

Table 1 Equal weight vs the three weighted/fine-grained directions

Direction	Voting object	New signal	New attack surface	Safety dependence	UDOS stance
P2 equal weight $2f+1$	final label	none	fewest	pure combinatorial (verified)	currently implemented
Direction one WBFT weighting	label + historical weight w_i	member credibility	weight manipulation, cold start	combinatorial $\times$ weight estimation	design direction, not implemented
Direction two COALITION-VAST	label + collusion detection	voter combination pattern	audit cost	combinatorial $\times$ collusion recognition	design direction, not implemented
Direction three BlockAgents PoT	each step of the reasoning chain	reasoning process	fan-in cost $O(\text{steps})$	combinatorial $\times$ process checking	design direction, not implemented

Figure 4 Trade-offs of equal weight and the three weighted/fine-grained directions

Figure 4 An overview of each direction's voting object, new signal, new attack surface, and safety dependence; each added signal increases both cheating cost and attack surface. Conceptual illustration.

The core judgment Table 1 conveys is: **each added signal (credibility, collusion pattern, reasoning process) is exchanged for a higher cheating cost while introducing a new attack surface and a new cost; safety degrades from "pure combinatorial" to "combinatorial $\times$ estimation quality of the new signal."** UDOS currently chooses to stop at the first row (verified by 10 tests) and registers the latter three rows as directions—this is not conservatism, but because P2's clean proof is precisely its most valuable asset.

6 Dross / Failure Boundaries / Falsifiable Conditions

Dross (what not to do):

1. **Do not use learned models to dynamically estimate weights on safety-critical paths.** H-CSC v2 has proven the learned encoder is defeated by the rule predicate in adversarial scenarios (AUROC 0.621–0.744 vs 0.865–0.982); weights must be computed with rule predicates (rolling agreement rate).
2. **Do not write "weighting is safer" as implemented.** UDOS has no weighted voting; any benefits are design hypotheses marked [RESULT NEEDED].
3. **Do not treat the percentages in user threads as facts.** The specific numbers of WBFT/CP-WBFT/COALITION-VAST/BlockAgents are all unverified in the v7.6.0 ledger and must carry "thread scope • unverified" when cited.

Failure boundaries:

- When historical weights $\$w_i\$$ are manipulated by "farm first, act maliciously later," the weighted committee loses defense against **lagging** malice—weight is a rear-view mirror that cannot see future betrayal;
- When colluders forge even the semantic core into alignment, "voting on process" and "voting on label" fail equally—PoT can only raise cheating cost, not eliminate collusion;
- The cost of weighted and fine-grained process voting grows with committee size and step count, and at very large scale may break through P1's fan-in budget.

Falsifiable conditions:

- If under the steelman baseline the wrong-stop rate of the rule-weighted committee is **not lower than** P2's equal-weight committee, the benefit hypothesis of direction one is overturned;
- If collusion recognition's recall on injected collusion attacks is not significantly higher than random guessing, the design principle of direction two does not hold;
- If the wrong-stop-rate improvement of voting on the reasoning chain is insufficient to cover its fan-in cost, direction three has no engineering value in the UDOS scenario.

7 Conclusion

P2's equal-weight $\$2f+1\$$ is a "minimal and clean" stop committee guarded by 10 property tests. This paper surveys its natural next dimension along three directions (credibility weighting, collusion governance, process consensus) but writes the conclusion as "**design direction, not implemented.**"

The reason is a simple but often-overlooked judgment: **how clean the proof of a safety mechanism is depends on how many external estimates it relies on; the reason equal-weight $2f+1$ can be written as a pure combinatorial proof is precisely that it assumes no member is more credible.** Once weights or process are introduced, safety becomes the product of "combinatorial $\times$ new-signal estimation quality"—and the H-CSC v2 withdrawal has just reminded us: that new-signal estimate may well, under a fair baseline, be worse than the simplest opponent.

First use the verified equal-weight committee steadily, then cautiously, step by step, introduce credibility signals with steelman contrasts; before then, weighting is a direction written in the design document, not code running in consensus.py.

References

Classical literature

1. Castro, M., & Liskov, B. (1999). Practical Byzantine Fault Tolerance. OSDI 1999. (the classical skeleton of equal-weight $n \geq 3f+1$, $2f+1$ quorum, view change; BFS only about 3% slower than unreplicated NFS)
2. Yin, M., Malkhi, D., Reiter, M. K., et al. (2019). HotStuff: BFT Consensus in the Lens of Blockchain. PODC 2019. arXiv:1803.05069.

2026 frontier threads (neither existence nor exact numbers verified in the v7.6.0 ledger; the following are user-research thread restatements, not facts)

1. Weighted BFT (WBFT class, IEEE TCCN direction): adaptively allocate voting weights by member response quality and credibility. [CITATION NEEDED: exact algorithm and numbers pending verification]
2. CP-WBFT (AAAI 2026 direction): user thread claims it can tolerate a high proportion of faults; claims such as "85.7% fault rate" unverified. [CITATION NEEDED]
3. COALITION-VAST (IEEE 2026-08 direction): coalition game theory + on-chain rule evolution + trust-driven targeted audit; user thread claims alignment 0.90 vs 0.69, coalition deviation -98%. [CITATION NEEDED: numbers pending verification]
4. BlockAgents (Shanghai Jiao Tong University, 2024 direction): Proof-of-Thought consensus + stake-based voter designation; user thread claims poisoning <3%, backdoor <5%. [CITATION NEEDED: numbers pending verification]

UDOS paper volume (evidence grades marked in text)

1. UDOS v7.5.0 P2 "BFT-lite for Multi-Agent Stop Decisions" (equal weight $n \geq 3f+1$, $q=2f+1$, 10 property tests verified; $n=7, f=2$, 24 orders cpu-`proto`).
 2. UDOS v7.6.0 P16 "Typed Finality", P32 "Rules over Learning" (sister papers; weighting/aggregation modules must pass steelman contrasts, weights use rule predicates).
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Evidence Discipline and Reproduction Notes

- This paper is a **design-direction registration** paper with no new experiment and no new implementation; UDOS's current `udos7/topology/consensus.py` only implements equal-weight BFT-lite and does not implement any weighting/collusion detection/process voting.

- The three recent threads (WBFT/CP-WBFT, COALITION-VAST, BlockAgents) all come from threads provided by user research, and their exact numbers have not been verified one by one in the v7.6.0 evidence ledger, all labeled "thread scope • unverified / [CITATION NEEDED]," and are not the factual basis of this paper's argument.
- The wrong-stop rate/rework rate/cost improvement of all weighting schemes are design hypotheses marked [RESULT NEEDED] ; if implemented in the future, they must first be extended on P2's 10 property-test suite and run through a steelman contrast (equal weight vs rule weighting), with ≥ 30 seeds before submission.
- P2's existing boundaries (SHA-256 hash placeholder signatures are not cryptographically secure, network partitions and adaptive attackers are not modeled, and semantic common-cause faults break through majority votes) **all hold and are more severe** for this paper's three directions—weighting/process voting does not eliminate common-cause faults, only raises cheating cost.